

Guangxu Yang (杨广绪)

PhD Student in Computer Science

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Location: Los Angeles, USA

Education

University of Southern California (USC)

Ph.D. Student in Computer Science

Advisor: Prof. Jiapeng Zhang

Los Angeles, USA

Aug 2022 – Current

University of Electronic Science and Technology of China (UESTC)

M.E. in Information and Communication Engineering

B.E. in Network Engineering

Chengdu, China

Sep 2019 – Jun 2022

Sep 2015 – Jun 2019

Research Interests

- Communication Complexity
- Quantum Algorithms

Publications

Note: Authors in theoretical computer science are listed alphabetically.

Conference Proceedings

- [1] **Exponential Separation of Quantum and Classical One-Way Numbers-on-Forehead Communication**
Guangxu Yang, Jiapeng Zhang.
67th Annual Symposium on Foundations of Computer Science (FOCS 2026).
- [2] **A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity**
Xudong Wu, Guangxu Yang, Penghui Yao.
The 53rd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2026).
- [3] **A Min-Entropy Approach to Multi-Party Communication Lower Bounds**
Mi-Ying Huang, Xinyu Mao, Shuo Wang, Guangxu Yang, Jiapeng Zhang.
40th Computational Complexity Conference (CCC 2025).
- [4] **Gadgetless Lifting Beats Round Elimination: Improved Lower Bounds for Pointer Chasing**
Xinyu Mao, Guangxu Yang, Jiapeng Zhang.
16th Innovations in Theoretical Computer Science (ITCS 2025).
- [5] **Communication Lower Bounds for Collision Problems via Density Increment Arguments**
Guangxu Yang and Jiapeng Zhang.
56th Annual ACM Symposium on Theory of Computing (STOC 2024).
- [6] **Communication Lower Bounds of Key-Agreement Protocols via Density Increment Arguments**
Mi-Ying Huang, Xinyu Mao, Guangxu Yang, Jiapeng Zhang.
21st Theory of Cryptography Conference (TCC 2023).

Preprints & Manuscripts

- [7] **Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication**
Guangxu Yang, Jiapeng Zhang. [arXiv:2506.12420](https://arxiv.org/abs/2506.12420).

- [8] **Does Approximate Compression Lead to Good Prediction? A Theory of Approximate MDL**
Qian Li, Xinyu Mao, Shang-Hua Teng, **Guangxu Yang**. [arXiv:2606.04834](#)
- [9] **Exponential Quantum Space Advantage for Approximating Max- k -SAT in the Streaming Setting**
Haoyu Wang, **Guangxu Yang**. [arXiv:2606.05366](#)
- [10] **Quantum Communication Lower Bounds for Search Problems from Matrix Discrepancy**
Minbo Gao, Chenghua Liu, **Guangxu Yang**, Tianyi Zhang.
- [11] **Exact Direct Sum Theorems for Shadow Circuits**
Xinyu Mao, **Guangxu Yang**.
- [12] **PRG Security and Collision Resistance of XOR Combiners in the Backdoored Random-Oracle Model**
Xinyu Mao, Shuo Wang, **Guangxu Yang**

Teaching Experience

Teaching Assistant at USC

- **CSCI 570:** Analysis of Algorithms *Spring 2026*
- **CSCI 170:** Discrete Methods in Computer Science *Spring 2025*
- **CSCI 575:** Quantum Computing and Quantum Cryptography *Spring 2024*

Professional Service

Conference Reviewer: MFCS 2026, FOCS 2026, PODS 2026, CRYPTO 2025, ITCS 2025, COCOON 2024.

Presentations

Conference Talks

Communication Lower Bounds of Key-Agreement Protocols

TCC 2023, Taipei Nov 2023

Gadgetless Lifting Beats Round Elimination

ITCS 2025, New York Jan 2025

Invited Seminar & Group Talks

Communication Lower Bounds for Collision Problems via Density Increment Arguments

University of Southern California Theory Lunch	Oct 2023
Nanjing University	Dec 2023
Shanghai Jiao Tong University	May 2024
Shanghai University of Finance and Economics	Oct 2025

Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication

University of Southern California Theory Lunch	May 2025
University of Science and Technology of China	Nov 2025
SoCal Theory Day at University of California San Diego	Feb 2026
Shanghai Jiao Tong University	Mar 2026